CS6140 Checkpoint: How Well Does Activity Recognition Work on New People?

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1. Changes since the proposal

Feedback noted that comparing a random split with a person split mixes window-overlap leakage with person shift, so I replaced it with a four-step split ladder:

- **A.** Random split, all windows.
- **B.** Random split, non-overlapping windows only (every other window).
- **C.** Same people, but tested on a later time block of each recording, with a one-window gap.
- **D.** Split by person.

A − B measures overlap, B − C closeness in time, and C − D person shift. I first verified the overlap: 95% of consecutive same-person, same-activity windows share exactly half their samples. Tuning now uses person-wise cross-validation on training people only.

2. Results so far

Experiment	LogReg	Random forest
Official split, tuned (paper's SVM: 96%)	95.0% (SVM: 95.6%)	—
Ladder with full training data: A → D	98.4% → 94.8%	97.9% → 93.2%
Ladder, equal training size: overlap / time / person (points)	0.0 / 2.3 / 0.7	0.4 / 2.2 / 0.7
Leave-one-person-out: mean (std), worst person	95.7% (4.6), 79.3%	93.4% (5.8), 78.6%
Learning curve: 2 → 21 training people	84.7% → 95.5%	80.7% → 92.5%

Other results: a model identifies the person from motion features with 58.2% accuracy on a random split and 49.1% on a time split (chance is 3.3%). A 1D-CNN on the raw signals reaches 93.6% on the official split.

3. What I learned

A random split overstates accuracy by about 4 points, and most of that comes from testing on moments close in time to training windows. Splits B and D train on the same number of windows (4,280), yet B scores 2.9 to 3.6 points higher.

Window overlap matters very little: removing it changed accuracy by 0 to 0.4 points at equal training size. With full data, A beats B by about 1 point, but A has twice the training data, and doubling B's data alone gives a similar gain.

A new person costs under 1 point on average, within the fold-to-fold spread of about 2 points, but individuals range from 79% to 100%, and person 14 is hardest for both models. People have a clear motion fingerprint (49% person-ID against 3.3% chance), yet the activity model barely suffers on new people. The main error is sitting predicted as standing (12%), and accuracy is still rising at 21 training people.

4. Problems and surprises

- The inflation (about 4 points) is far below the 10 to 16 points in Dehghani et al. (2019), possibly because UCI HAR was recorded in a lab with scripted activities.
- Matching training sizes by subsampling removed most of split A's overlapping neighbors, so I also ran the ladder with full training sets.
- The CNN peaked at epoch 15, then overfit; I report the final epoch (93.6%).
- Person-wise tuning lowered logistic regression (95.5% to 95.0%); I report the tuned value rather than choosing by test accuracy.

5. Remaining work before Dec 8

- Repeat the split ladder with several random seeds to put error bars on the three gaps.
- Choose the CNN's epoch count using held-out training people, and average over 3 seeds.
- Find out why person 14 is hard: which activities fail for that person.
- Write the final report.